

Home Credit Default Risk

Model Development, Statistical Comparison, Probability Calibration, and Risk Diagnostics

This report is generated from the current full notebook run. It evaluates 5 classification models on 307,511 public application records using mutually exclusive development partitions.

Development recommendation: retain XGBoost as the observed discrimination leader and LightGBM as the principal challenger. The two leading final-test AUC intervals overlap. The pre-specified DeLong comparison is not statistically significant at 5%. No production champion is approved from this experiment alone.

Model	AUC	KS	PR-AUC	F1	Raw Brier	Final fit (s)
XGBoost	0.7674	0.4087	0.2609	0.3196	0.1925	4.63
LightGBM	0.7671	0.4038	0.2593	0.3195	0.1945	2.34
Random Forest	0.7526	0.3801	0.2364	0.2983	0.1748	41.32
Logistic Regression	0.7514	0.3848	0.2286	0.2907	0.2048	6.03
Decision Tree	0.7231	0.3281	0.2081	0.2737	0.2115	2.35

Report status

- Notebook integrity checks: all passed.
- Captured fit warnings requiring action: 0.
- Use classification: model-development evidence on a public benchmark dataset.
- Excluded claims: out-of-time stability, reject inference, long-run PD calibration, expected loss, profitability, and regulatory capital adequacy.

Prepared from reproducible output artifacts
Evidence directory: outputs/ (relative to the repository root)

1. Executive Summary

The highest observed final-test AUC is 0.7674 for XGBoost, with a stratified bootstrap 95% interval of 0.7589 to 0.7765. LightGBM follows at 0.7671, with an interval of 0.7584 to 0.7765. These are observed random-holdout results, not evidence that either algorithm will dominate across future booking vintages.

The pre-specified paired DeLong comparison uses LightGBM and XGBoost, the two candidates chosen by model-selection AUC. The AUC difference is -0.0003 with $p=0.6445$. It is not statistically distinguishable at the 5% level under this conditional test.

Class weighting changes the effective event prior used during training. Raw model outputs are therefore treated as ranking scores rather than accepted portfolio PDs. Prior correction, Platt scaling, and Isotonic regression are assessed on an independent calibration partition, with calibration-in-the-large, slope, observed-to-expected ratio, Brier score, and Log Loss reported on the final test.

A model selection decision should balance discrimination, calibration, stability, explainability, operating cost, implementation effort, and governance. This report names development candidates only; a production champion still requires out-of-time testing and portfolio-specific economics.

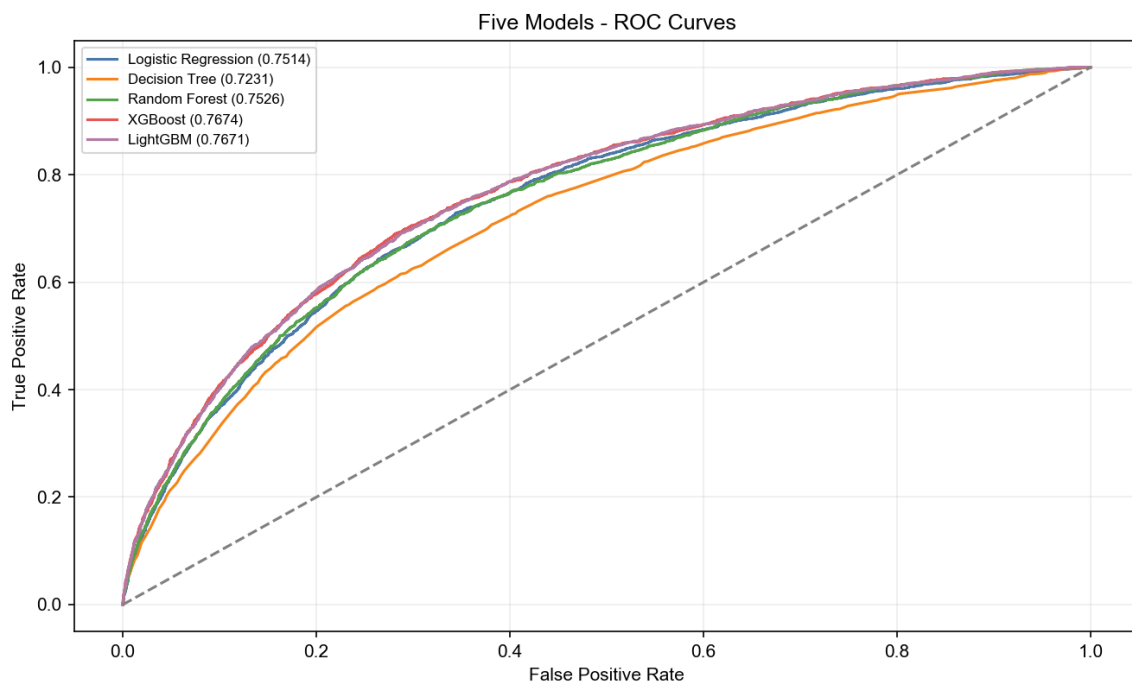


Figure 1. Final-test ROC curves for all five models.

2. Data, Target, and Experimental Design

2.1 Data scope

The source is Kaggle's Home Credit Default Risk application table. The current run reads 307,511 rows and 122 source columns. TARGET=1 denotes repayment difficulty in the competition label; it is not a legal bankruptcy, charge-off, or institution-specific Basel default definition.

2.2 Independent data roles

The design separates model fitting, boosted-model selection, probability calibration, operating-threshold selection, and final evaluation. Feature filtering and preprocessing are fitted on training data only.

Split	Rows	Bad Count	Bad Rate
train	184506	14895	0.0807
model_selection	30751	2482	0.0807
calibration	30751	2483	0.0807
threshold	30751	2482	0.0807
test	30752	2483	0.0807

Temporal limitation: the public application table does not provide a reliable booking timestamp for the target population. The final partition is a stratified random holdout, not an out-of-time sample.

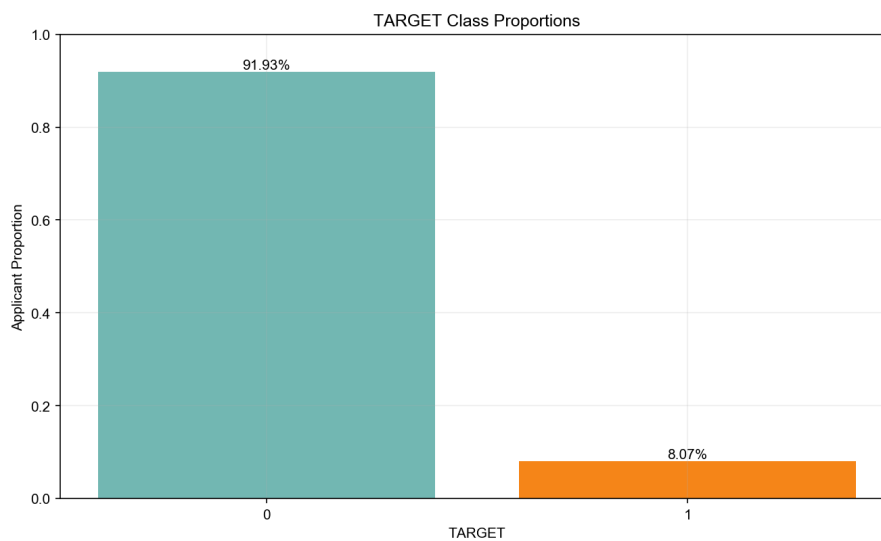


Figure 2. Observed target rate in the development data.

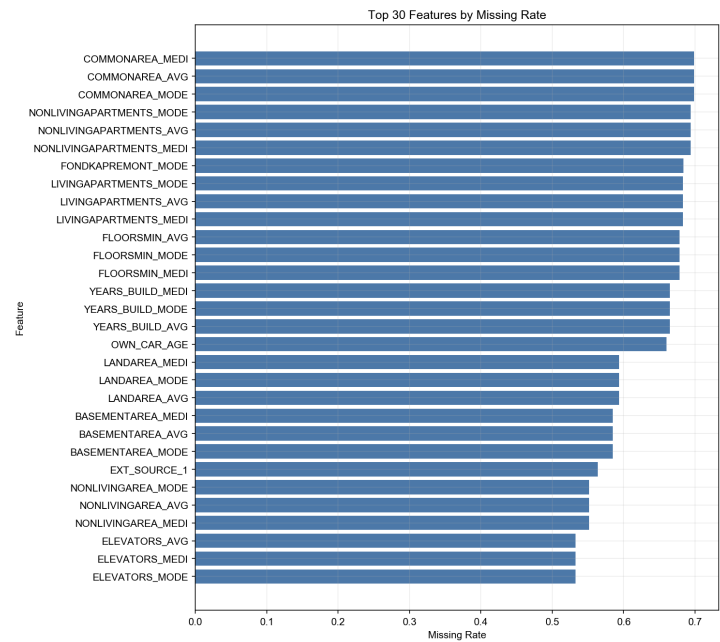


Figure 3. Highest source-feature missing rates.

3. Data Preparation and Feature Controls

The notebook replaces known sentinel day values, standardizes non-finite values, retains missingness for pipeline imputation, and constructs a small set of business-ratio variables. Ratios use protected division and do not silently preserve infinite values.

Training-only screening retains 108 raw and engineered columns and produces 232 encoded model inputs. The process removes near-constant, extremely incomplete, and highly correlated columns using training statistics only.

3.1 Selected feature sample

Selected Feature
NAME_CONTRACT_TYPE
CODE_GENDER
FLAG_OWN_CAR
FLAG_OWN_REALTY
CNT_CHILDREN
AMT_INCOME_TOTAL
AMT_CREDIT
AMT_ANNUITY
NAME_TYPE_SUITE
NAME_INCOME_TYPE
NAME_EDUCATION_TYPE
NAME_FAMILY_STATUS
NAME_HOUSING_TYPE
REGION_POPULATION_RELATIVE
DAYS_BIRTH
DAYS_EMPLOYED
DAYS_REGISTRATION
DAYS_ID_PUBLISH
OWN_CAR_AGE
FLAG_MOBIL

3.2 Exclusion evidence

The audit table records 19 excluded columns and their reasons. TARGET and SK_ID_CURR are explicitly excluded from model inputs. The complete lists remain available as CSV evidence rather than being summarized only in narrative text.

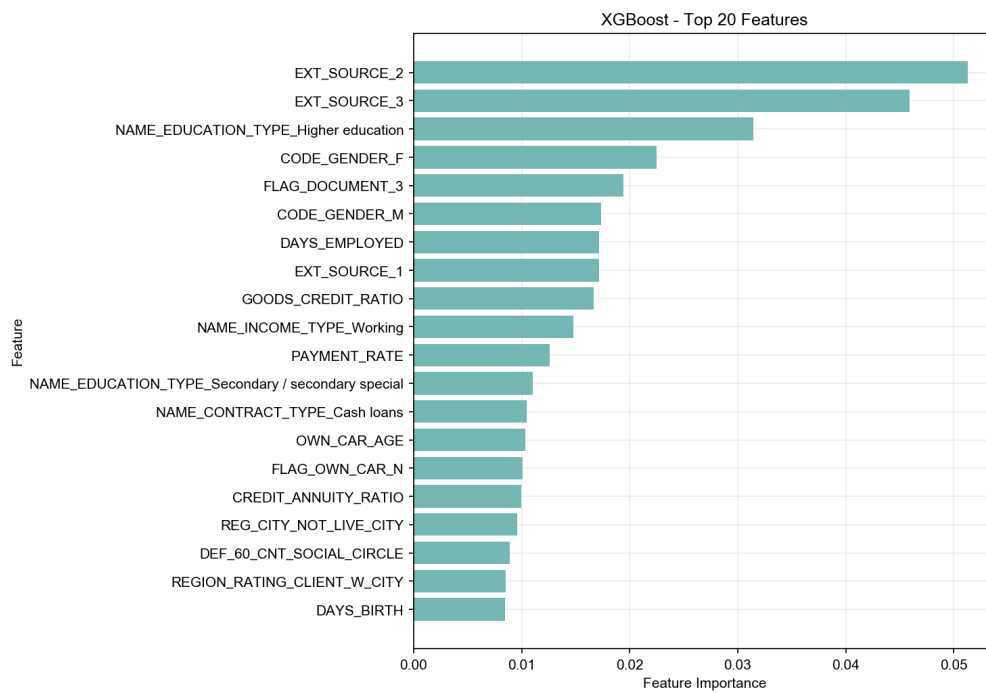


Figure 4. Built-in feature importance for XGBoost.

4. Model Development and Parameter Screening

Five model families are compared: Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM. The search is deliberately described as limited parameter screening. Candidate counts and timing are read from the run, rather than inferred from the size of a nominal parameter space.

Model	Candidates	Search AUC	Screening (s)	Final fit (s)	Boundary reached
Logistic Regression	6	0.7400	39.87	6.03	
Decision Tree	8	0.7054	13.00	2.35	
Random Forest	8	0.7386	541.09	41.32	
XGBoost	8	0.7554	1341.31	4.63	False
LightGBM	8	0.7574	17.99	2.34	False

4.1 Comparable timing definition

Every reported final training time measures one clean fit on the complete training partition after parameter selection. XGBoost and LightGBM use early stopping only during candidate screening, then refit the selected iteration count on all training rows. This aligns the timing definition with the scikit-learn models.

4.2 LightGBM row sampling

LightGBM sets `subsample_freq=1`, so candidate values below `subsample=1` activate row bagging. The notebook checks this parameter in the saved final model evidence.

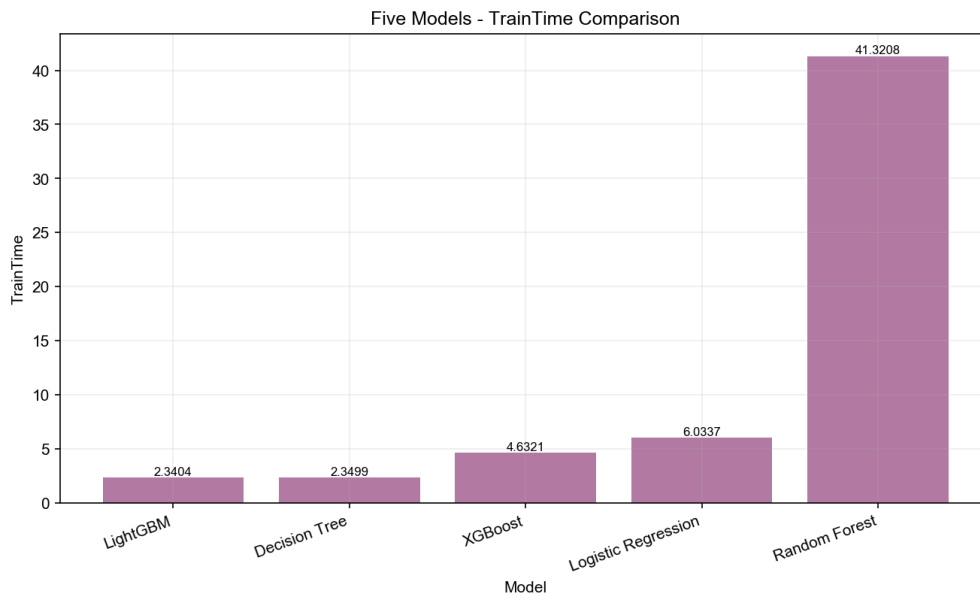


Figure 5. Clean final-fit time by model.

5. Final-Test Performance and Uncertainty

AUC, Accuracy Ratio, KS, and PR-AUC assess ranking. F1 and related threshold metrics use the maximum-F1 cutoff selected only on the separate threshold partition. Brier score and Log Loss shown here refer to raw model outputs and should be interpreted alongside the calibration section.

Model	AUC	AR	KS	PR-AUC	Precision	Recall	F1	Brier	Log Loss
XGBoost	0.7674	0.5348	0.4087	0.2609	0.2517	0.4374	0.3196	0.1925	0.5663
LightGBM	0.7671	0.5341	0.4038	0.2593	0.2394	0.4801	0.3195	0.1945	0.5696
Random Forest	0.7526	0.5052	0.3801	0.2364	0.2324	0.4160	0.2983	0.1748	0.5321
Logistic Regression	0.7514	0.5028	0.3848	0.2286	0.2294	0.3967	0.2907	0.2048	0.5976
Decision Tree	0.7231	0.4462	0.3281	0.2081	0.1922	0.4752	0.2737	0.2115	0.6083

5.1 Bootstrap confidence intervals

Model	AUC	Lower 95%	Upper 95%	Bootstrap SE	Replicates
XGBoost	0.7674	0.7589	0.7765	0.0048	500
LightGBM	0.7671	0.7584	0.7765	0.0048	500
Random Forest	0.7526	0.7434	0.7615	0.0050	500
Logistic Regression	0.7514	0.7428	0.7604	0.0048	500
Decision Tree	0.7231	0.7126	0.7330	0.0053	500

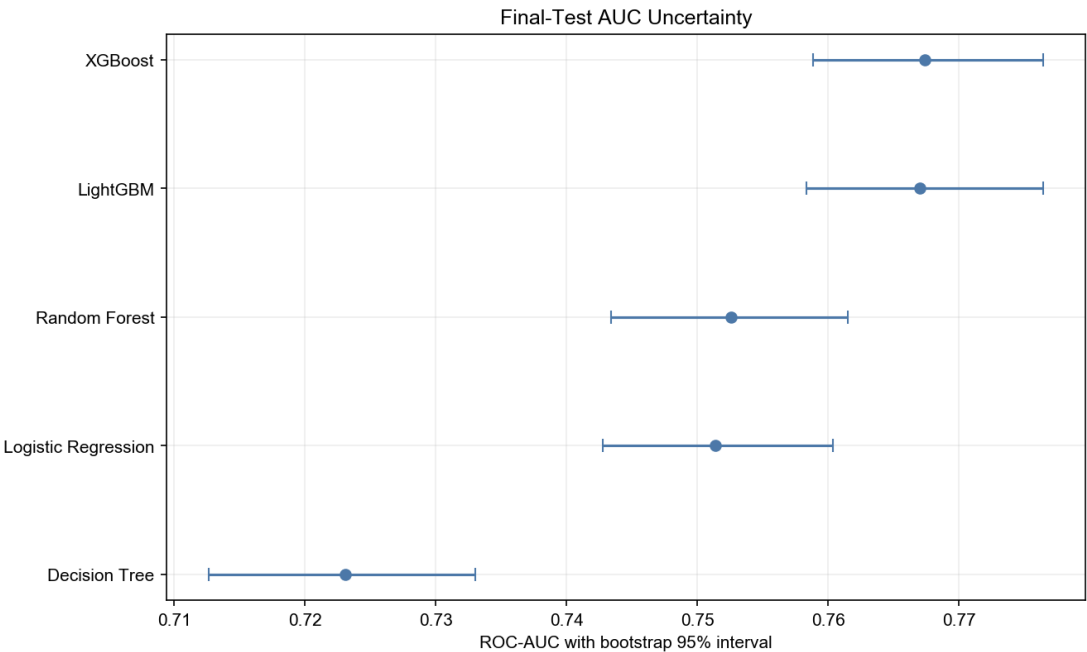


Figure 6. Stratified bootstrap AUC intervals on the final random holdout.

6. Paired Comparison and Generalization Diagnostics

6.1 Pre-specified DeLong comparison

Model A	Model B	AUC A	AUC B	Difference	Lower 95%	Upper 95%	Two-sided p
LightGBM	XGBoost	0.7671	0.7674	-0.0003	-0.0018	0.0011	0.6445

The pair is determined by model-selection AUC before inspecting final-test ranking. Paired bootstrap results for every model pair are retained in `paired_bootstrap_auc_differences.csv`. A p-value does not measure business materiality or future-vintage stability.

6.2 Train-to-model-selection gaps

Model	Train AUC	Selection AUC	Test AUC	Train-selection gap
Random Forest	0.8518	0.7470	0.7526	0.1049
XGBoost	0.7919	0.7639	0.7674	0.0280
LightGBM	0.7875	0.7646	0.7671	0.0229
Decision Tree	0.7372	0.7190	0.7231	0.0182
Logistic Regression	0.7461	0.7474	0.7514	-0.0013

These gaps are continuous diagnostics. The report does not use an arbitrary fixed cutoff to label a model as overfitted. Ranking stability still requires repeated development samples and out-of-time evidence.

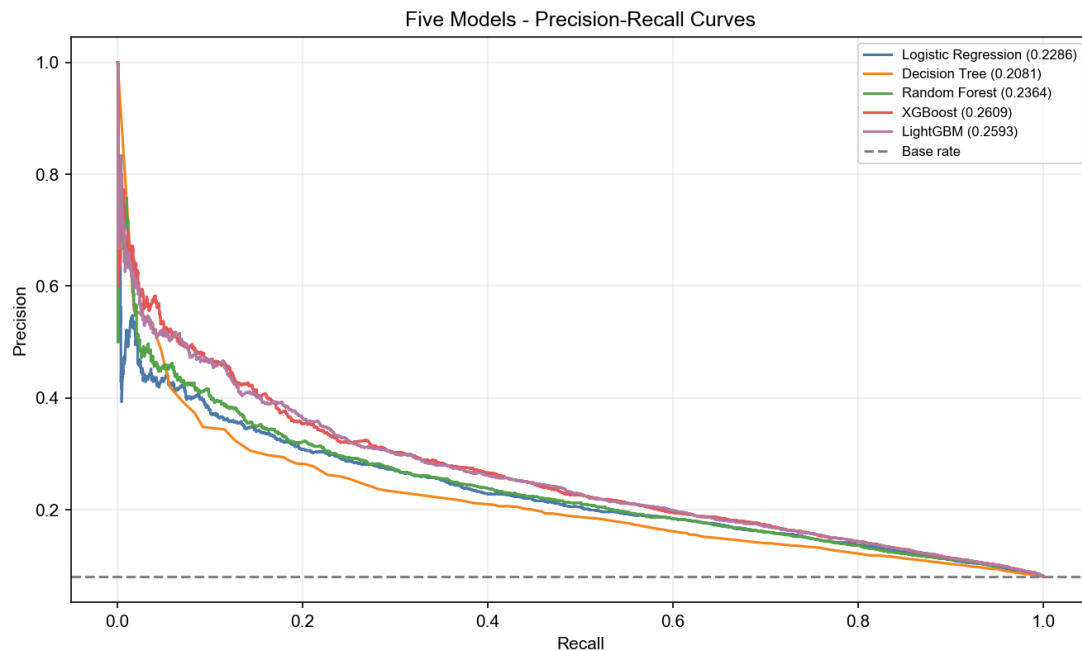


Figure 7. Precision-recall curves under class imbalance.

7. Risk Ranking and Decile Diagnostics

Risk deciles test whether higher predicted scores concentrate observed bad outcomes. They are descriptive on the final random holdout and should not be interpreted as stable production grades.

Model	Adjacent increase rate	Top-decile bad capture	Top-decile lift
XGBoost	1.000	35.079%	3.51
LightGBM	1.000	34.636%	3.46
Random Forest	1.000	32.420%	3.24
Logistic Regression	1.000	32.380%	3.24
Decision Tree	1.000	29.118%	2.91

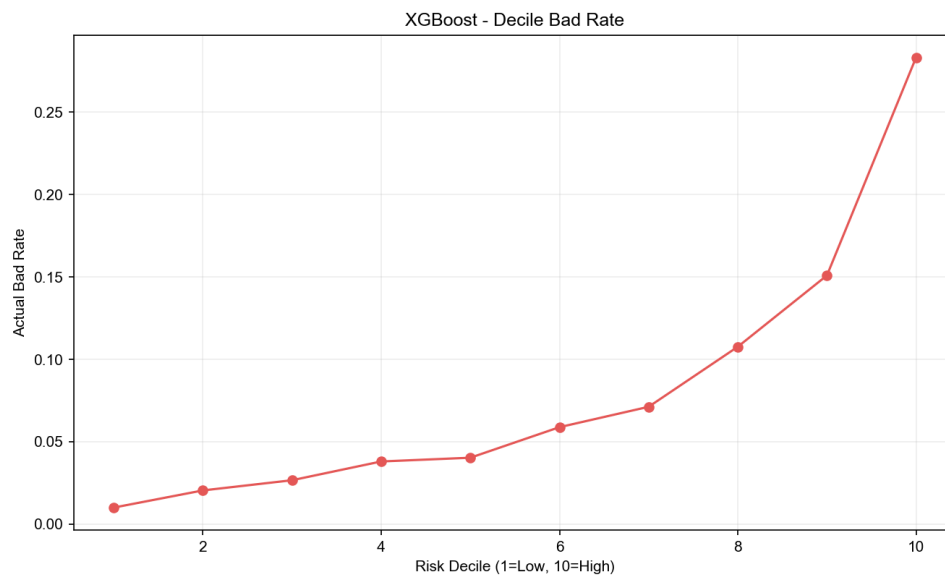


Figure 8. XGBoost observed bad rate by risk decile.

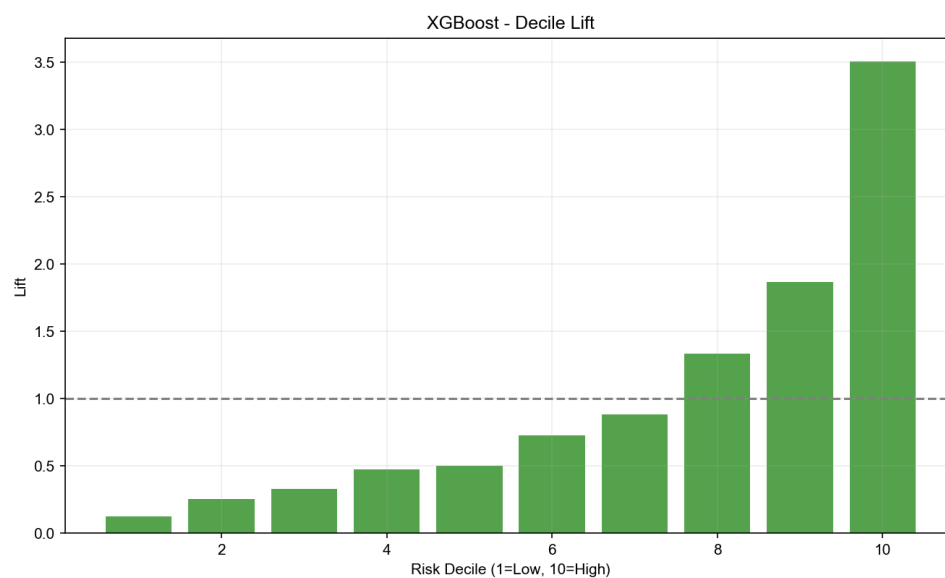


Figure 9. XGBoost lift by risk decile.

8. Probability Calibration and PD Interpretation

Balanced class weights and scale_pos_weight alter the effective event prevalence seen by the learner.

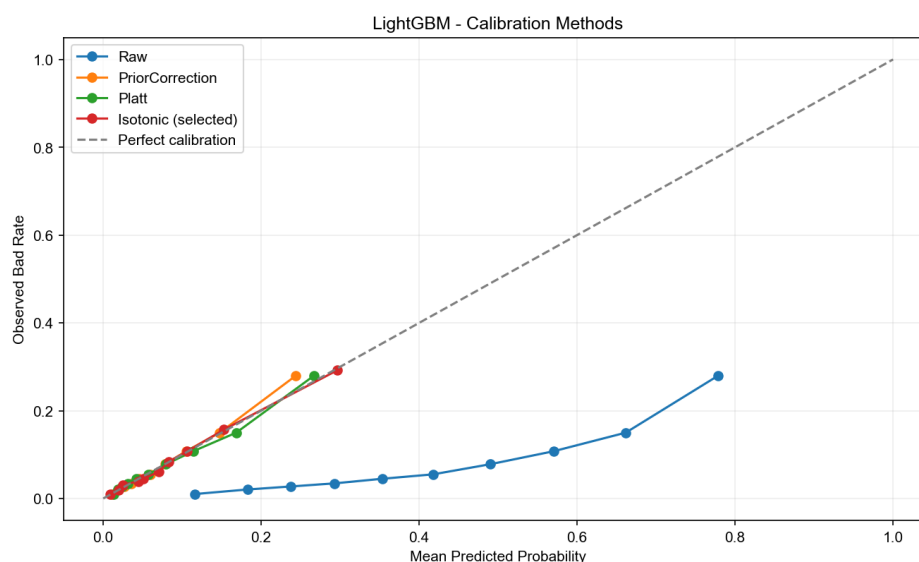
Consequently, raw predict_proba output is not assumed to represent the unadjusted portfolio PD. The project compares analytical prior correction, Platt scaling, and Isotonic regression on a separate calibration sample.

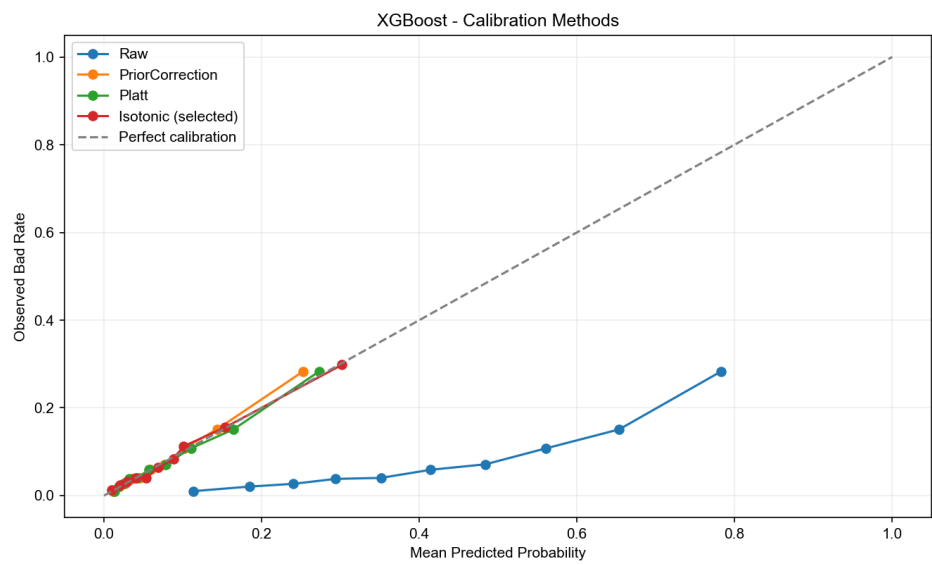
Model	Selected calibration	Calibration selection brier	Brier	Logloss
LightGBM	Isotonic	0.0679	0.0673	0.2449
XGBoost	Isotonic	0.0679	0.0672	0.2461

8.1 Final-test calibration diagnostics

Model	Method	Observed	Mean PD	CITL	Slope	O/E	Brier
LightGBM	Raw	0.0807	0.4099	-2.3796	1.1057	0.1970	0.1945
LightGBM	PriorCorrection	0.0807	0.0772	0.0528	1.1067	1.0462	0.0674
LightGBM	Platt	0.0807	0.0814	-0.0098	1.0175	0.9918	0.0674
LightGBM	Isotonic	0.0807	0.0815	-0.0107	0.9581	0.9912	0.0673
XGBoost	Raw	0.0807	0.4078	-2.3772	1.0834	0.1980	0.1925
XGBoost	PriorCorrection	0.0807	0.0770	0.0553	1.0851	1.0481	0.0672
XGBoost	Platt	0.0807	0.0813	-0.0085	1.0169	0.9929	0.0673
XGBoost	Isotonic	0.0807	0.0814	-0.0100	0.9347	0.9917	0.0672

PD boundary: the sample bad rate is a point-in-time public-data rate, not a through-the-cycle long-run default rate. No long-run portfolio anchor is available for intercept anchoring, grade-level calibration, or regulatory PD estimation.





Calibration comparison for XGBoost.

9. Segment Performance and Random-Split PSI

Segment diagnostics cover gender, contract type, age band, and income band where event counts are sufficient. Ranges flag heterogeneity for review; they do not prove discrimination or unfairness because composition, sample size, and confidence intervals differ by segment.

Segment variable	Model	Minimum AUC	Maximum AUC	Segments
AgeBand	Decision Tree	0.6495	0.7364	5
AgeBand	LightGBM	0.6919	0.7807	5
AgeBand	Logistic Regression	0.6598	0.7641	5
AgeBand	Random Forest	0.6898	0.7620	5
AgeBand	XGBoost	0.6865	0.7796	5
ContractType	Decision Tree	0.6929	0.7242	2
ContractType	LightGBM	0.7296	0.7689	2
ContractType	Logistic Regression	0.7231	0.7524	2
ContractType	Random Forest	0.7240	0.7534	2
ContractType	XGBoost	0.7337	0.7689	2
Gender	Decision Tree	0.7120	0.7186	2
Gender	LightGBM	0.7605	0.7624	2
Gender	Logistic Regression	0.7413	0.7478	2
Gender	Random Forest	0.7430	0.7497	2
Gender	XGBoost	0.7617	0.7621	2
IncomeQuartile	Decision Tree	0.7193	0.7333	4
IncomeQuartile	LightGBM	0.7563	0.7781	4
IncomeQuartile	Logistic Regression	0.7386	0.7670	4
IncomeQuartile	Random Forest	0.7430	0.7632	4
IncomeQuartile	XGBoost	0.7554	0.7799	4

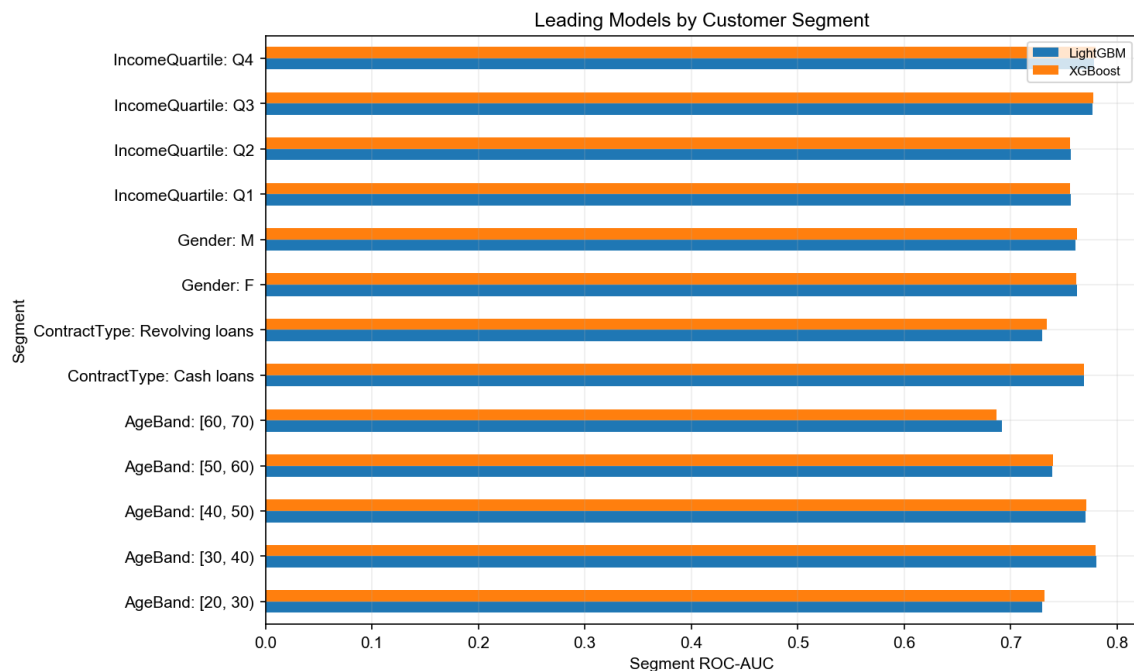


Figure 10. Segment AUC for leading development candidates.

The maximum random-test-versus-training PSI is 0.0018. This is a pipeline consistency diagnostic under a random split. It must not be presented as evidence of temporal population stability.

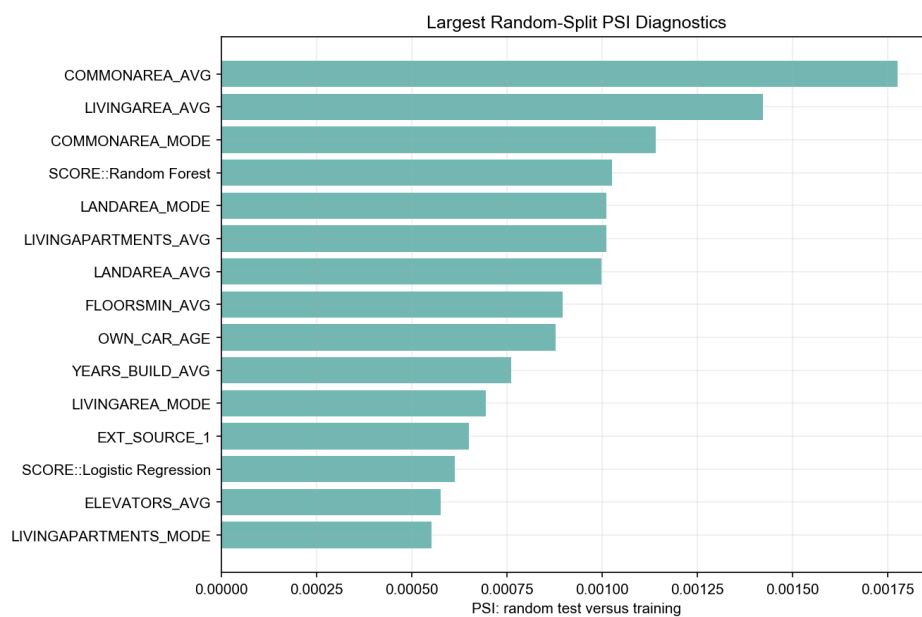


Figure 11. Highest random-split PSI values.

10. Illustrative Decision Costs

Threshold selection depends on the relative cost of approving a bad applicant and rejecting a good applicant. The table below uses stated cost ratios only. It does not estimate expected credit loss, revenue, funding cost, operating expense, customer lifetime value, or capital consumption.

Model	Bad:good cost	Threshold	Reject rate	Bad capture	Cost / applicant
LightGBM	1.0	0.4109	1.46%	8.50%	0.0816
LightGBM	5.0	0.1538	15.51%	46.68%	0.3327
LightGBM	10.0	0.0922	32.44%	69.39%	0.5155
XGBoost	1.0	0.4160	1.40%	8.38%	0.0812
XGBoost	5.0	0.1749	11.01%	37.62%	0.3316
XGBoost	10.0	0.0924	32.38%	69.79%	0.5113

11. Model Choice: What Can Be Defended

Development recommendation: retain XGBoost as the observed discrimination leader and LightGBM as the principal challenger. The two leading final-test AUC intervals overlap. The pre-specified DeLong comparison is not statistically significant at 5%. No production champion is approved from this experiment alone.

Why retain XGBoost

- It has the highest observed final-test AUC (0.7674) and KS (0.4087) in this run.
- Its ranking evidence is supported by bootstrap intervals, paired comparisons, decile diagnostics, and segment tables rather than a single point estimate.
- Its final fit and inference artifacts reproduce saved predictions under the notebook checks.

Why not declare the alternatives inferior

- LightGBM: observed AUC is close (0.7671); it remains a credible challenger where runtime, implementation, or calibration is preferable.
- Logistic Regression: weaker nonlinear capacity, but the strongest transparency benchmark and often easier to govern, explain, and monitor.
- Random Forest: can fit richer interactions, but a larger train-selection gap and heavier explanation burden may not be justified by its holdout gain.
- Decision Tree: highly readable rules, but the single-tree structure has lower discrimination, coarser probability steps, and higher partition instability.

The final choice should therefore be framed as a conditional development recommendation. A lender may reasonably choose a slightly lower-AUC model when explainability, monitoring simplicity, latency, or model-risk policy outweighs a small and uncertain discrimination difference.

12. Limitations and Required Next Evidence

- Out-of-time validation: unavailable because the application table lacks a reliable target-population booking timeline.
- Repeated development samples: not yet run; one split cannot establish ranking stability across random seeds.
- Reject inference: not performed because rejected-applicant outcomes and the acceptance mechanism are unavailable.
- Portfolio PD anchoring: no long-run default rate supports TTC calibration or intercept anchoring.
- Economic and capital analysis: LGD, EAD, margin, cost, funding, capacity, and capital inputs are absent.
- Historical-table aggregation: bureau, previous-application, and repayment tables are not implemented and are not presented as completed functionality.
- Interpretability: coefficients and built-in feature importance are descriptive; SHAP analysis is not claimed or silently substituted.

13. Reproducibility and Audit Trail

The notebook validates disjoint identities, finite metrics, prediction-file AUC reproduction, saved-model reload agreement, feature leakage controls, active LightGBM bagging, required evidence files, model artifacts, and coverage of bootstrap and segment results.

Validation check	Passed
five_models_trained	True
development_splits_are_disjoint	True
all_core_metrics_finite	True
saved_predictions_reproduce_auc	True
reloaded_models_reproduce_predictions	True
no_target_or_id_in_features	True
lightgbm_subsample_frequency_active	True
no_critical_fit_warnings	True
required_evidence_files_exist	True
exact_expected_model_artifacts	True
bootstrap_intervals_cover_all_models	True
segment_results_cover_all_models	True

13.1 Warning policy

Warnings are captured around fitting and saved to warnings_log.csv; they are not globally hidden. The current run records 0 warning(s). Parameters reported by a library as ignored or ineffective, and convergence warnings, fail the notebook validation check.

13.2 Evidence files

Core evidence includes model_metrics.csv, test_predictions.csv, best_parameters.csv, candidate-level boosting results, bootstrap_metric_intervals.csv, paired_bootstrap_auc_differences.csv, delong_auc_test.csv, calibration_diagnostics.csv, segment_performance.csv, random_split_psi.csv, cost_sensitivity.csv, saved models, and this dynamically generated report.

14. Conclusion and Production Approval Gates

Development recommendation: retain XGBoost as the observed discrimination leader and LightGBM as the principal challenger. The two leading final-test AUC intervals overlap. The pre-specified DeLong comparison is not statistically significant at 5%. No production champion is approved from this experiment alone.

XGBoost is the observed leader in this run, but the evidence does not support a universal or production-ready superiority claim. The defensible outcome is a governed challenger set, with final approval deferred until temporal, stability, calibration, explainability, and economic evidence is available.

Approval gate	Current status	Evidence required before production
Temporal stability	Open	Out-of-time sample across representative booking vintages
Ranking stability	Open	Repeated development splits and confidence on rank changes
Portfolio PD calibration	Development only	Current portfolio rate and long-run default anchor
Segment review	Preliminary	Larger samples, uncertainty, policy and fairness review
Decision economics	Illustrative only	LGD, EAD, margin, costs, capacity, and capital
Explainability	Partial	Reason-code validation and adverse-action review
Independent validation	Open	Reperformance, challenge, and governance approval

This conclusion is intentionally narrower than a formal institutional model validation opinion. It states what the current experiment supports, what remains unresolved, and which evidence would change the decision.